

# Measurement and Instrumentation (EE-214)



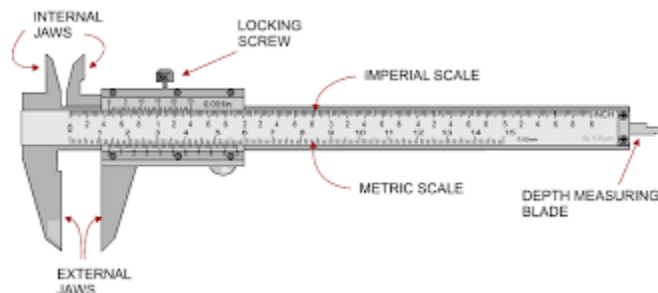
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# Description of Content in Brief:

1. Significance and methods, Classification of Instruments, Static Characteristics, Source of errors.
2. Moving coil, Moving Iron, Electro dynamometer, Rectifier type instruments, Measurement of Power and Energy, Ohmmeters, Megger.
3. Transducers- Resistance, Inductance and Capacitance transducers. Measurement of Displacement, Speed, Liquid level, Pressure, Temperature, Strain. Hall Effect Devices.
4. A.C and D.C. Bridges –Measurement of resistance, inductance and capacitance.
5. Instrument Transformers- Current and Potential transformers.

# Introduction

- Measurement of a given entity is an act or the result of **comparison** between the **unknown quantity** (whose magnitude is unknown) and a **predefined standard**.
- The result of comparison is expressed in terms of **numerical value**.
- The measurement is the process by which one can **convert physical parameters in to meaningful numbers**.
- The numerical measure must have **unit** for meaningful result.



- There are two basic **requirements** of measurement-
  - The **measurement standard** used for comparison purpose must be **accurately defined**.
  - The **apparatus** used and method adopted must be provable.
- **Instruments:** An instrument is a device to measure a physical quantity such that temperature, pressure or electrical parameters like voltage, resistance etc.

OR

- **Device** which is used to **measure the unknown quantity**.
- **Instrumentation**-It is an art or science which is used to measure the physical quantity

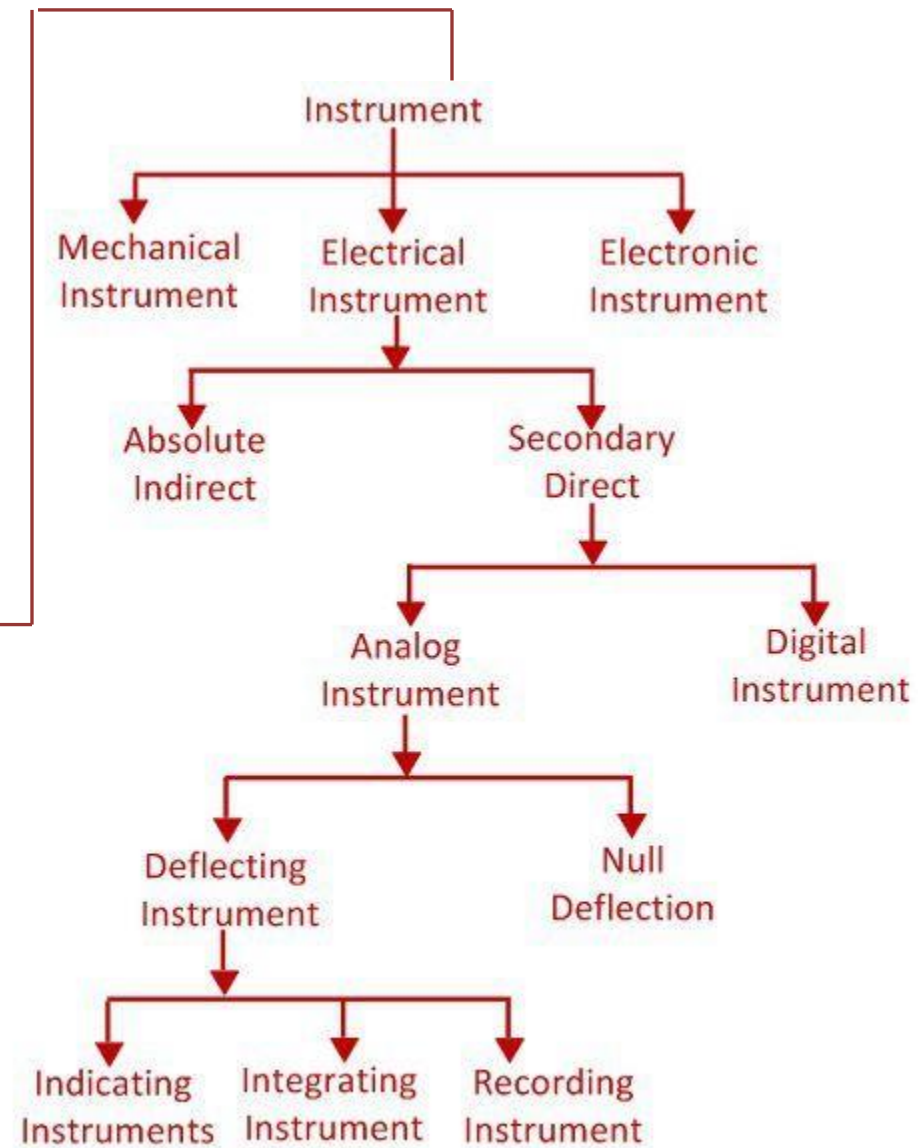
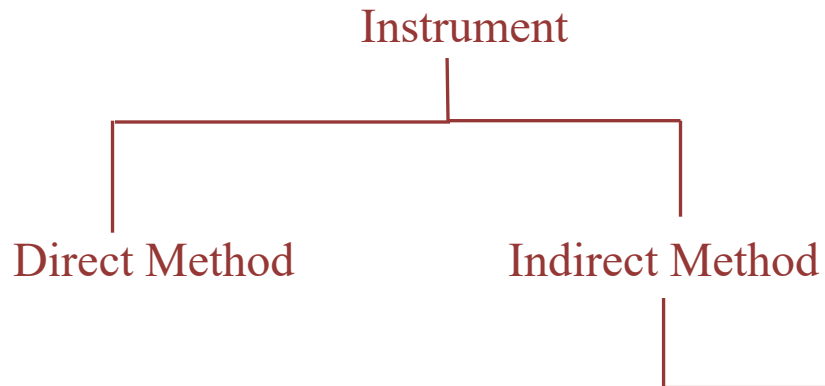


Voltmeter



Speedometer

# Classification of Instruments



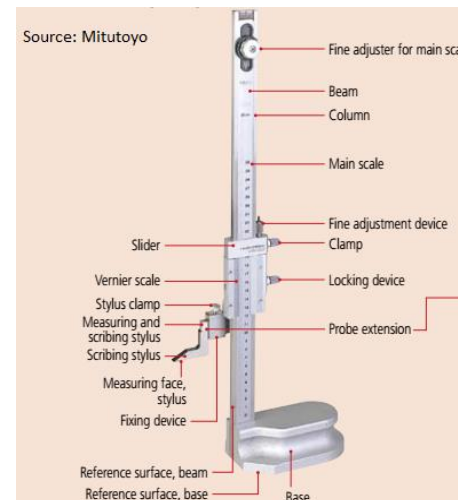
- **Direct Method**-In this method, **unknown quantity** is directly compared against a **standard** (Ex. Scale, Ruler etc).
- The result is expressed in terms **number and unit**.
- Direct methods are quite common for measurement of *physical quantities like length and mass*.
- Measurement by direct methods are not always *possible, feasible and practical*.
- It is less **sensitive** and **inaccurate**.



- **Indirect Method**-In this method, unknown quantity (Ex. Voltage, Current, Frequency, Temperature, speed etc.) is indirectly measured.
- In **engineering applications**, measuring instruments are being designed using **indirect method**.

- *Indirect method* based instruments are divided in *three categories*-
  - **Mechanical Instrument**
  - **Electrical Instrument**
  - **Electronics Instrument**

- **Mechanical Instrument**-In this method, result is obtained by moving or setting the parts of instrument.
- **Mechanical instruments** are very **reliable for static and stable condition**.
- They are **unable to respond rapidly** to measure the **dynamic and transient condition** because they are having moving parts.
- It is **not possible** to measure **50 Hz voltage** or **current** due to heavy mass of moving parts.
- Examples: **Vernier Calliper, Vernier height gauge, Bevel Protractor etc.**





- **Electrical Instruments-**
- Electrical Instruments consist of a coil and permanent magnet.
- When current flows in the coil, torque is produced, **pointer moves** on the **calibrated scale**.
- **Adv.-**Electrical instruments **indicate** the output **more rapidly** than mechanical instruments.
- **Disadv.-**There is **movement of pointer** so it is **not too fast** as electronic instruments.
- **Voltmeter, Ammeter, wattmeter etc.**

- **Electronic Instrument**-In this method, **magnitude of unknown quantity** is obtained due to the **motion of electrons**.
- It can capture the **very fast response** (transient and dynamic conditions of few ms and  $\mu$ s) which is the requirement of scientific and industrial measurement.
- Frequency Counter, Multi-meter, Cathode ray Oscilloscope (CRO) etc.



# Electrical Instruments

- **Absolute Instrument**-This measurement gives the **magnitude or result** of the unknown quantity under measurement in **terms of physical constant and its deflection**.
- Ex. **Tangent Galvanometer**- Measurand current is to be calculated from tangent of angle of deflection produced, horizontal component of earth's magnetic field, radius and no. of turns of coil.



$$I = \frac{2aB_h}{\mu_o n} \tan \theta$$

a- radius of Coil, n= Number of turns of coil,  
 $B_h$  = Horizontal component of Earth's  
magnetic field ( **$1.8 \times 10^{-5} \text{ W/m}^2$** )

$\mu_o$  = Permeability of free space ( **$4\pi \times 10^{-7}$** )  
H/m

Tangent galvanometer

- **Secondary Instrument**-It is classified into two categories-

- Analog Instruments
- Digital Instruments



- **Analog Type-** If pointer continuous moves on the calibrated scale then it is called analog instrument. Examples are ‘**Permanent Magnet Moving Coil instrument (PMMC)**’, Needle type **speedometer** of automobile, **Moving Iron (MI), Mercury thermometer.**
- **Digital Type-** If output is in digits form then it is called digital instrument. Examples are **Digital multi-meter**, **Digital Blood Pressure measuring instruments**, **Digital watches** etc.

- **Advantages of Digital Instrument-**

- Reading or display is in digital form therefore errors involve due to human being is eliminated.
- The reading may be taken to any number of significant figure or number.
- Output may be directly stored in memory because it is digital form.
- Power requirement is very small.
- Measurement of low level signal is possible.
- High Frequency Range.
- Accuracy is more.
- Better Resolution (*The smallest increment of measurement that a instrument is capable of making*) .

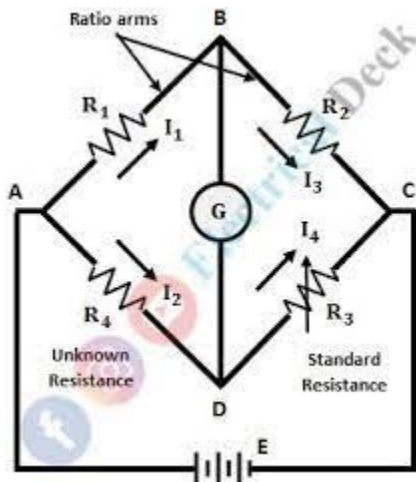
- **Disadvantages of Digital Instrument-**

- Their construction is complex.
- These are expensive devices.
- There operation is affected by temperature.
- Digital instruments are mechanically weak.
- Their repair and maintenance cost are high
- Instrument shows discontinuous operation.

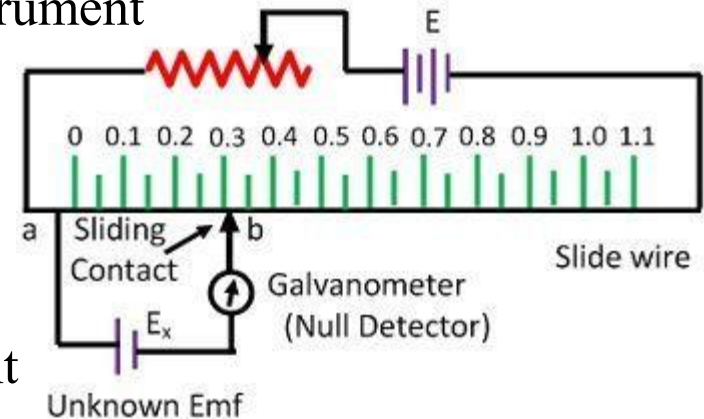
- **Analog Instruments** are divided into two categories-
  - **Deflecting Instruments:** In this type of instrument, pointer deflects or moves on the calibrated scale. Examples- Voltmeter, Ammeter etc
  - **Null Type Instruments:** In this type of instrument, pointer shows zero deflection. Examples- Wheatstone Bridge, Null Type Galvanometer



Deflecting Type of Instrument



Null Type of Instrument



# Deflecting Instruments

- Deflecting instruments are classified into three categories:
  - **Indicating Instruments.**
  - **Integrating Instruments.**
  - **Recording Instruments.**

# Indicating Instruments

- This type of instrument will indicate or show the result of measuring quantities only at a particular instant.



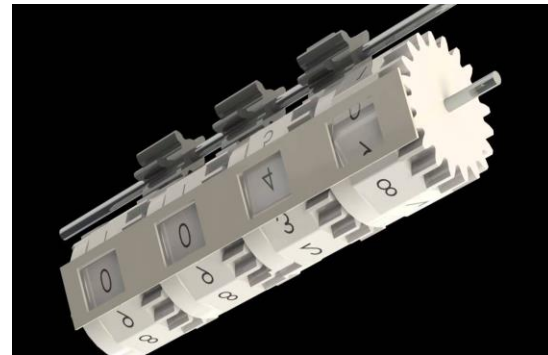
- It cannot store any previous values.
- The indication of the measuring values is done with a pointer and scale in case of analog meters.
- In case of digital circuits, LCD (liquid crystal display) or LED (light-emitting diode) glows for a moment.
- Examples of indicating instruments are voltmeters, ammeters, speedometers, multimeter etc.



# Integrating Instruments

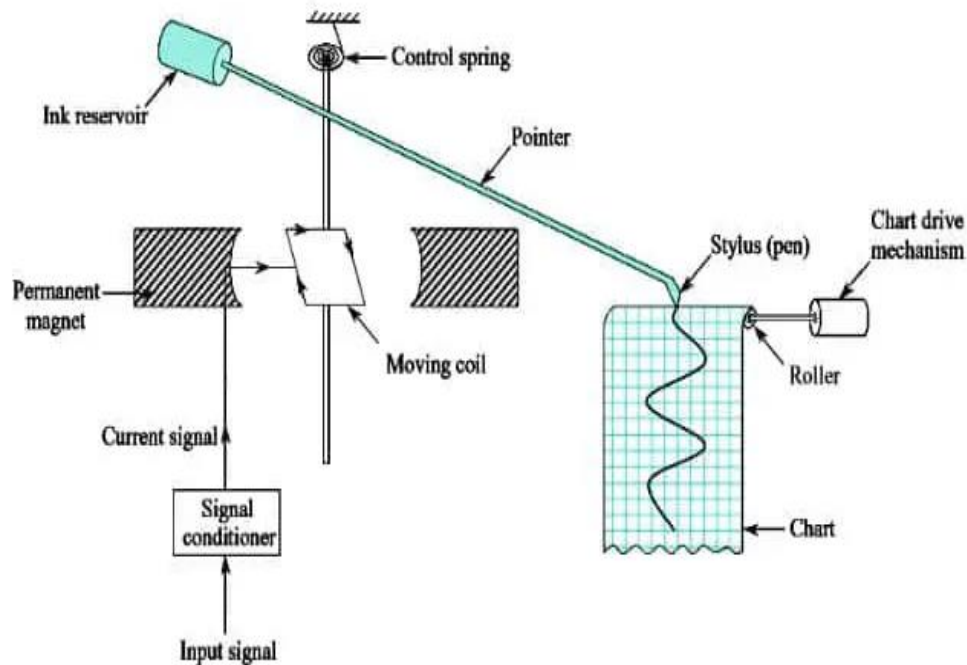
- These instruments give the integration of the inputs applied over a particular period of time.
- There is a special arrangement of gears that drives the rings on which the numbers from 0 to 9 are marked in case of analog meters. Examples of these instruments are odometers, watt-hour (Energy) meters.

In the case of digital meters, a electronic counter is used to integrate the inputs over the periods of time. Output of counter is given to the display unit. Examples are display for waiting (Restaurants), traffic lights etc.



# Recording Instruments

- The **recording instrument** is defined as the instrument which records the continuous variation of the **unknown quantity magnitude** for a **particular period**.



- The **ink pen** is **moved** for recording the readings on the **graph paper**.
- The pen is continuously rotated on the graph at the constant speed. The path traced by the pen gives a continuous reading of the physical quantity. Exp. ECG.